

Absolute Market Unfairness, Liquidity Premia, and the Compression of Put–Call Parity Violations in Cryptocurrency Option Markets

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Abstract

This paper introduces *Absolute Market Unfairness* (AMU), a model-free measure of the positive arbitrage profits implied by quoted put–call-parity deviations after contract-specific conversion and trading-cost adjustments. The empirical setting is the inverse BTCUSD and ETHUSD option markets of Deribit and OKX, observed from 2020 through early 2026, where contract design, collateral denomination, and settlement conventions prevent a naive transfer of textbook put–call parity formulas. We show how AMU can be computed from synchronized option and futures quotes in directly comparable basis-point units, so that cross-exchange, cross-underlying, and cross-segment comparisons are meaningful. We then document how AMU evolves over time and across the option surface: AMU is lower for BTC than for ETH, lower near the money than away from the money, lower in longer-dated contracts than in short-dated ones, and higher during periods of market stress. Finally, we interpret these patterns through a limits-to-arbitrage framework in which observed parity wedges persist inside an endogenous no-arbitrage band generated by spreads, depth constraints, stale quotes, funding frictions, and institutional frictions. In this way, AMU serves both as a transparent market-quality measure and as an empirical proxy for the cost of enforcing the law of one price in segmented crypto derivative markets.

Keywords: cryptocurrency options, put-call parity, liquidity premium, limits to arbitrage, market efficiency, ETF approval, MiCA

JEL classification: G12, G14, G15, G23

1 Introduction

Put–call parity (PCP) is one of the most basic no-arbitrage restrictions in option pricing. In a frictionless market, a European call and put with the same strike and maturity must differ in value by the forward value of the underlying. In actual markets, however, parity holds only inside a band. Bid–ask spreads, fees, margin requirements, collateral

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segmentation, stale quotes, and operational frictions all affect whether an apparent wedge can be monetized.

This paper studies that wedge in cryptocurrency option markets. Our starting point is *Absolute Market Unfairness* (AMU), a model-free measure of the positive arbitrage profit implied by quoted option and futures prices after contract-specific conversion and explicit trading-cost adjustments. AMU is designed for crypto derivatives markets, where inverse settlement, non-standard collateral conventions, and fragmented exchange design make textbook parity relations insufficient on their own. By expressing the executable wedge in basis points of comparable notional, AMU turns heterogeneous option quotes into a common market-quality measure.

The trading costs are knowable, but they are specified in a very complex way. Throughout this paper we assume a fixed proportional cost of 0.003 for a full arbitrage round trip. This is reasonable but definitely not exact. However, the time-dependence of the AMU measure is robust to the exact cost level. Different venues can also be compared directly from the point of view of efficiency if the same cost structure is assumed. At the end of the day, the trading cost assumptions only serve to ensure that AMU values are reasonable, but the comparisons do not depend on them.

The paper then asks a second question: what economic object does AMU measure? The cross-sectional evidence already points to a liquidity-premium interpretation. AMU is higher for ETH than for BTC, higher away from the money than near the money, higher in short maturities than in longer maturities, and higher during periods of market stress. Those are exactly the segments in which arbitrage is harder to implement and market makers can preserve wider effective markups. This suggests that observed PCP deviations are not best understood as random pricing mistakes, but as the residual compensation left to liquidity suppliers when the law of one price is costly to enforce.

The time-series motivation is the institutionalization of crypto markets around 2024. The approval of U.S. spot Bitcoin exchange-traded products in January 2024, the later Ether ETF milestones, and the staged implementation of MiCA plausibly expanded the set of investors able to deploy arbitrage capital in a legally and operationally robust way (U.S. Securities and Exchange Commission 2024^{b,a}, European Securities and Markets Authority 2024). Our working hypothesis is therefore not that traders suddenly discovered parity, but that the no-arbitrage band narrowed as custody, regulation, execution, and competition improved.

The contribution of the paper is threefold. First, we define AMU in a way that is implementable for inverse crypto options and matching futures. Second, we document the main empirical regularities of AMU in a 2020–2026 sample of Deribit and OKX BTCUSD and ETHUSD options. Third, we interpret the resulting wedges through a limits-to-arbitrage framework in which changes in market structure affect the cost of parity enforcement.

The remainder of the paper is organized as follows. Section 2 reviews the PCP and liquidity literature. Section 3 defines AMU, explains its implementation, and summarizes the main empirical patterns. Section 4 presents the liquidity-premium interpretation and threshold view of parity enforcement. Section 5 describes the panel data, institutional event dates, and empirical specifications. Section 6 concludes.

2 Related literature

2.1 *Put–call parity and crypto derivative efficiency*

Testing put–call parity has a long tradition in equity, index, and currency option markets. Early work shows that apparent parity violations are sensitive to bid–ask spreads, execution timing, short-sale constraints, taxes, and transaction costs (Klemkosky & Resnick 1979, Nisbet 1992, Kamara & Miller 1995, El-Mekkaoui & Flood 1998, Cremers & Weinbaum 2010). Crypto option markets create additional complications because contracts are often inverse or quanto-style instruments, collateral and settlement currencies may differ from payoff currencies, and the underlying reference index is not always directly tradable (Alexander et al. 2023, Lucic & Sepp 2024).

Recent crypto work increasingly treats derivative efficiency as a joint market-structure problem. Alexander et al. (2024) develop arbitrage tests linking Bitcoin and Ether options with perpetual futures and find that crypto derivative markets have become more efficient over time, especially for longer-maturity options. Felföldi-Szűcs (2024) study high-frequency put–call parity in Binance Ether options and emphasize that violations become much smaller once synchronization and trading costs are handled carefully. The present paper extends this line of work to inverse options and matching futures on OKX and Deribit by introducing a directly comparable measure of investor-facing arbitrage profits in basis points.

2.2 *Liquidity premia and the limits of arbitrage*

The liquidity-premium interpretation of AMU is motivated by the idea that arbitrage is costly and capital constrained. Amihud & Mendelson (1986) show that bid–ask spreads affect asset pricing because investors require compensation for illiquidity. Shleifer & Vishny (1997) emphasize that textbook arbitrage is typically not riskless or capital-free in practice. Gârleanu & Pedersen (2011) formalize how margin requirements and funding constraints generate deviations from the law of one price. These ideas are directly relevant to crypto derivatives, where collateral denomination, margin rules, exchange credit risk, and stablecoin conversion all affect the cost of enforcing parity.

The closest mature-market analogy is covered interest parity (CIP). In frictionless international money markets, CIP should bind tightly. Nevertheless, post-crisis bank balance-sheet constraints and dollar funding pressures generated persistent CIP deviations (Du et al. 2018). The crypto case studied here is the mirror image: regulatory clarity and institutional access may have reduced, rather than increased, the effective cost of arbitrage capital, leading to a compression of parity deviations.

ETF pricing provides a second analogy. ETF prices are kept close to net asset value by authorized participants who create and redeem shares. However, when the underlying assets are illiquid or arbitrageurs face balance-sheet constraints, ETF premiums and discounts can persist (Pan & Zeng 2017, Todorov 2021). The 2024 introduction of U.S. spot Bitcoin ETFs is particularly relevant because it created a regulated arbitrage and custody channel between traditional brokerage infrastructure and digital asset exposure.

3 Absolute Market Unfairness: definition, implementation, and baseline facts

3.1 Why AMU is needed in crypto options

Put–call parity remains the right no-arbitrage benchmark in crypto options, but the implementation is contract-specific. The most liquid contracts in our sample are inverse BTCUSD and ETHUSD options on Deribit and OKX, where contracts are quoted against a USD index but collateral, premium, and settlement are denominated in the underlying cryptocurrency. As a result, parity must be written after converting option premia and futures legs into a common currency using the platform-specific payoff convention (Alexander et al. 2023, Lucic & Sepp 2024). We therefore focus on inverse options and matching expiry futures, rather than presenting a general taxonomy of all crypto contract classes in the main text.

MMA is path-specific, because executable profit depends on forward/backward implementation and on whether the call-side or put-side quote is joined. The four path definitions are shown in Table 1. The positive part of the MMA distribution is then aggregated into AMU, which is a daily summary measure of the residual arbitrage wedge after contract-specific conversion and trading-cost adjustments.

Table 1: Textbook arbitrage profit formulas by path and contract style. To obtain the basis-point statistics used for MMA/AMU, multiply each expression by 10,000/index_level. Each expression includes the trading-cost term $\text{cost} = \text{cost_per_notional} (0.003) \times \text{index_level}$ from Table 4. Contract size ends up not mattering in the basis-point expression for MMA. For inverse contracts, S_{bid} and S_{ask} represent the cost of converting the coin payout to USD or stablecoin.

Path	Linear contracts	Inverse contracts
fwd, join call	$C_{bid} - P_{bid} - (F_{bid} - K)e^{-rT} - \text{cost}$	$(C_{bid} - P_{bid})S_{bid} - (F_{bid} - K)e^{-rT} - \text{cost}$
fwd, join put	$C_{ask} - P_{ask} - (F_{bid} - K)e^{-rT} - \text{cost}$	$(C_{ask} - P_{ask})S_{ask} - (F_{bid} - K)e^{-rT} - \text{cost}$
bck, join call	$P_{ask} - C_{ask} + (F_{ask} - K)e^{-rT} - \text{cost}$	$(P_{ask} - C_{ask})S_{ask} + (F_{ask} - K)e^{-rT} - \text{cost}$
bck, join put	$P_{bid} - C_{bid} + (F_{ask} - K)e^{-rT} - \text{cost}$	$(P_{bid} - C_{bid})S_{bid} + (F_{ask} - K)e^{-rT} - \text{cost}$

3.2 From executable arbitrage to AMU

Let $X_{i,t}$ denote the quoted put–call parity wedge for option pair i at time t , expressed in basis points of the underlying notional after contract-specific conversion into a common currency. In a standard textbook setting, $X_{i,t} = 0$. In our terminology, the signed path-level quantity is *market maker arbitrage* (MMA):

$$\text{MMA}_{i,t} = \frac{\Pi_{i,t}}{N_{i,t}} \times 10,000, \quad (1)$$

and AMU is the positive-part average of MMA:

$$\text{AMU}_{g,t} = \frac{1}{n_{g,t}} \sum_{i \in g,t} \max(0, \text{MMA}_{i,t}), \quad (2)$$

where $\Pi_{i,t}$ is the best available PCP-arbitrage profit across the implementable arbitrage paths and $N_{i,t}$ is the option contract’s underlying notional. In practice, $\Pi_{i,t}$ is computed

from synchronized option and futures quotes using the relevant inverse-contract payoff structure (Alexander et al. 2023, Lucic & Sepp 2024), with explicit commission and bid–ask adjustments. The path-level linear and inverse formulas are summarized in Table 1.

For inverse crypto contracts, option premia are first converted into a common settlement unit via index scaling, consistent with the inverse payoff conventions in Alexander et al. (2023), Lucic & Sepp (2024). Operationally, we evaluate these executable paths on synchronized quotes, convert each path into basis points of comparable notional (this is the MMA), and then aggregate with $\max(0, \text{MMA})$ to obtain AMU. The daily AMU summary is therefore not a settlement-price anomaly. It is the average positive executable wedge left in quoted prices after the contract mechanics and trading frictions are handled correctly.

3.3 Contract-specific implementation

For inverse options on OKX and Deribit, option premia are quoted in the underlying cryptocurrency while strikes and reference prices are stated in USD index terms. The parity comparison must therefore combine option premia, matching expiry futures, and the appropriate conversion between the option payout unit and the futures settlement unit. The empirical implementation follows three rules. First, put and call quotes are synchronized with the corresponding expiry futures quote. Second, parity is evaluated only for matched strike–expiry pairs with usable bid and ask prices. Third, the implied MMA profit of each executable path is converted into a notional-scaled basis-point quantity so that BTC and ETH, and OKX and Deribit, can be compared directly.

This scaling is central. Raw dollar profits would mechanically make deeper markets look worse because a small proportional wedge can still correspond to a larger absolute dollar amount. AMU instead measures the proportional investor-facing inefficiency of the quoted market.

3.4 Data and implementation

The empirical sample uses high-frequency Deribit and OKX BTCUSD and ETHUSD inverse options and matching expiry futures from Tardis. The panel spans 2020 through early 2026. Observations are synchronized on a regular time grid, filtered to matched put–call pairs with the same strike and expiry, and aggregated into daily cells indexed by exchange, underlying, relative-strike bucket, and time-to-expiry bucket. The resulting panel preserves the cross-section needed to compare liquidity states while also supporting time-series analysis of AMU. Table 2 summarizes the coverage of the matched option sample that underlies the AMU calculation.

3.5 Baseline AMU patterns

Before turning to the liquidity-premium interpretation, the AMU measure yields four baseline empirical facts that organize the rest of the paper. First, AMU is systematically lower for BTC than for ETH. Second, AMU is lowest near the money and rises in less liquid strike regions. Third, AMU is typically lower for longer-dated contracts than for the shortest maturities. Fourth, AMU spikes during periods of market stress, when market making becomes balance-sheet intensive and quote quality deteriorates.

These are not causal claims by themselves. They are descriptive regularities that establish what the liquidity-premium explanation must account for. They also show why

	BTCUSD		ETHUSD	
	deribit	okex	deribit	okex
Avg num contracts	109	83	67	52
Avg num expirations	7	6	6	4
Avg num strikes	80	57	53	46
Avg call spread (bp)	46	35	46	36
Avg put spread (bp)	42	29	53	32
Num days in sample	366	366	366	366

Table 2: Summary statistics of matched put–call pairs in the 2020–2026 sample.

AMU is more informative than raw spreads alone: it isolates the part of the quoted spread structure that survives the full parity-arbitrage construction.

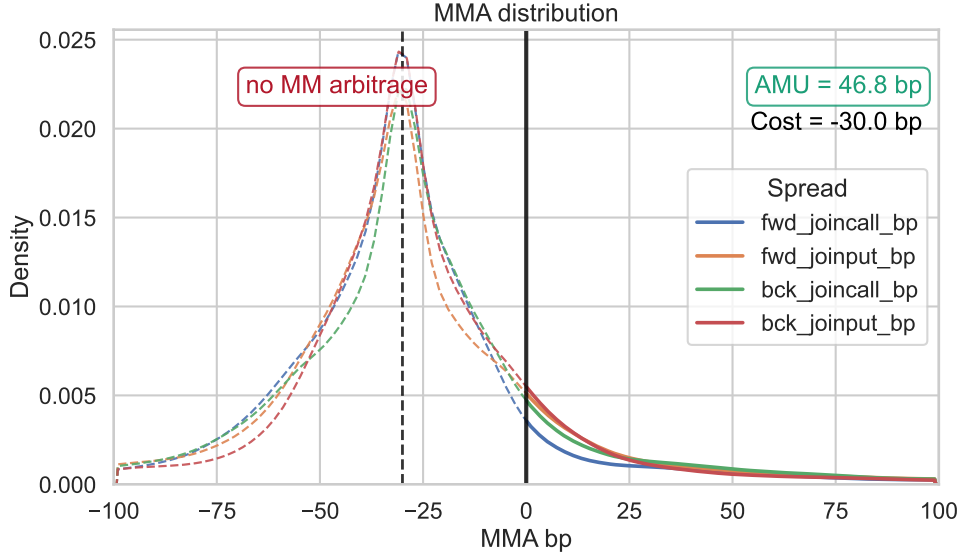


Figure 1: Average distribution of the four MMA paths across all markets.

Figure 1 reports the average path densities across all markets, and Figure 2 compares the total MMA histogram for Pre/Post 2024. These are the distributional objects behind the AMU summary measure.

4 AMU as a liquidity premium

We interpret AMU as a lower bound on the liquidity premium associated with the enforcement of parity. It captures the part of the quoted bid–ask structure that leaves a risk-free profit to a market maker who passively joins the top of the order book and completes the arbitrage cycle once filled. Thus AMU is not merely a pricing error. It is a measure of how much compensation the marginal liquidity supplier can extract because competitors do not immediately close the spread.

The no-arbitrage restriction should therefore be written as a band:

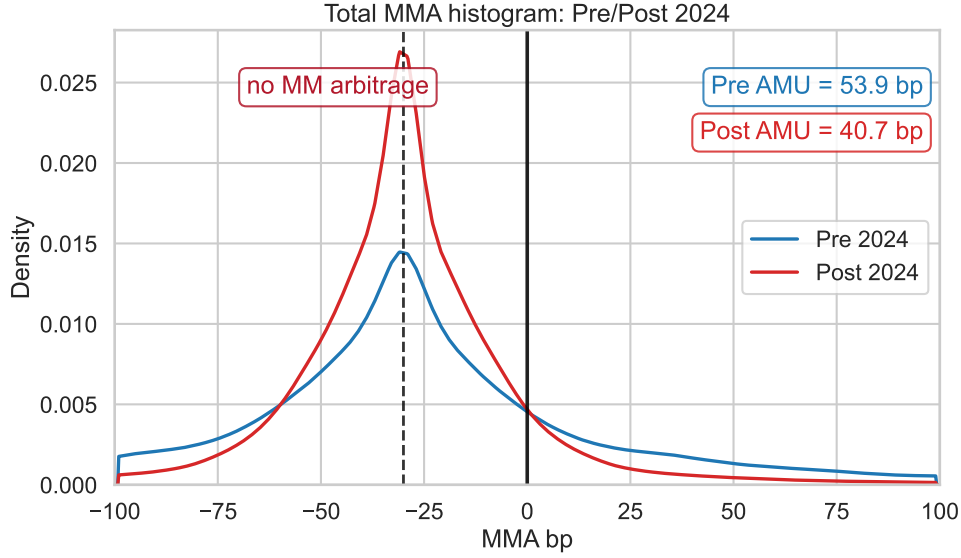


Figure 2: Total MMA histogram across all four paths, split into Pre/Post 2024 periods.

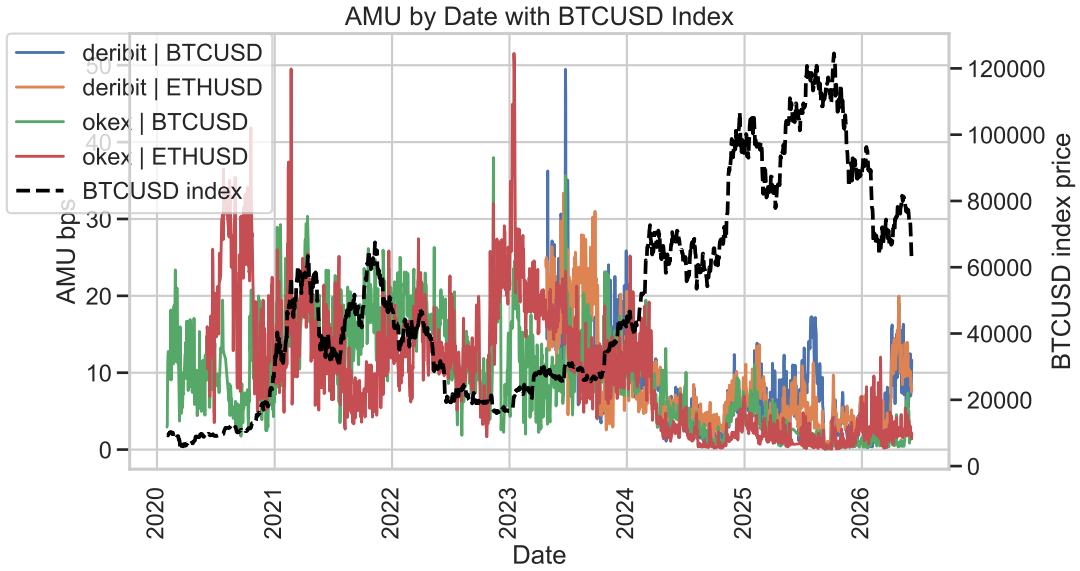


Figure 3: Daily AMU by exchange and underlying in the 2020–2026 sample.

		mean AMU(bps)	positive MMA occurs	Num Observations
underlying	exchange			
BTCUSD	deribit	5.34	4,852,286	11,541,389
	okex	4.98	3,084,146	8,765,581
ETHUSD	deribit	5.44	3,174,854	7,056,276
	okex	3.56	1,617,190	5,508,204

Table 3: Global summary results across exchanges and underlyings.

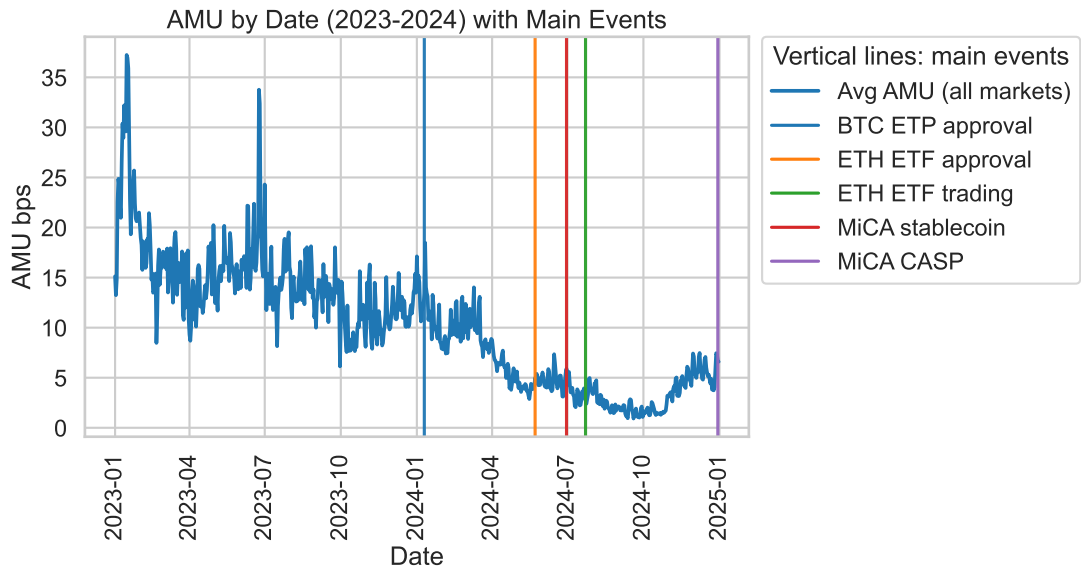


Figure 4: Daily average AMU across all markets from 2023-01 through 2024-12, with main institutional event dates marked by vertical lines.

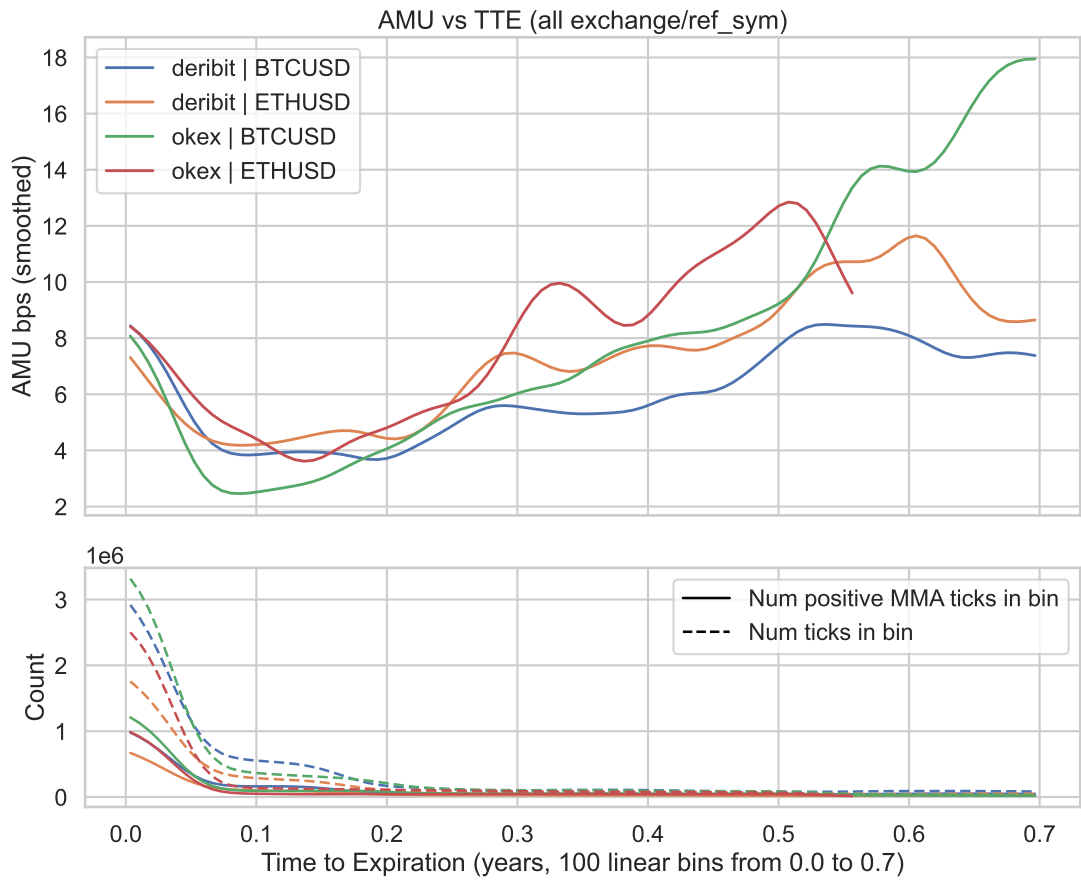


Figure 5: AMU by time to expiry across exchanges and underlyings.

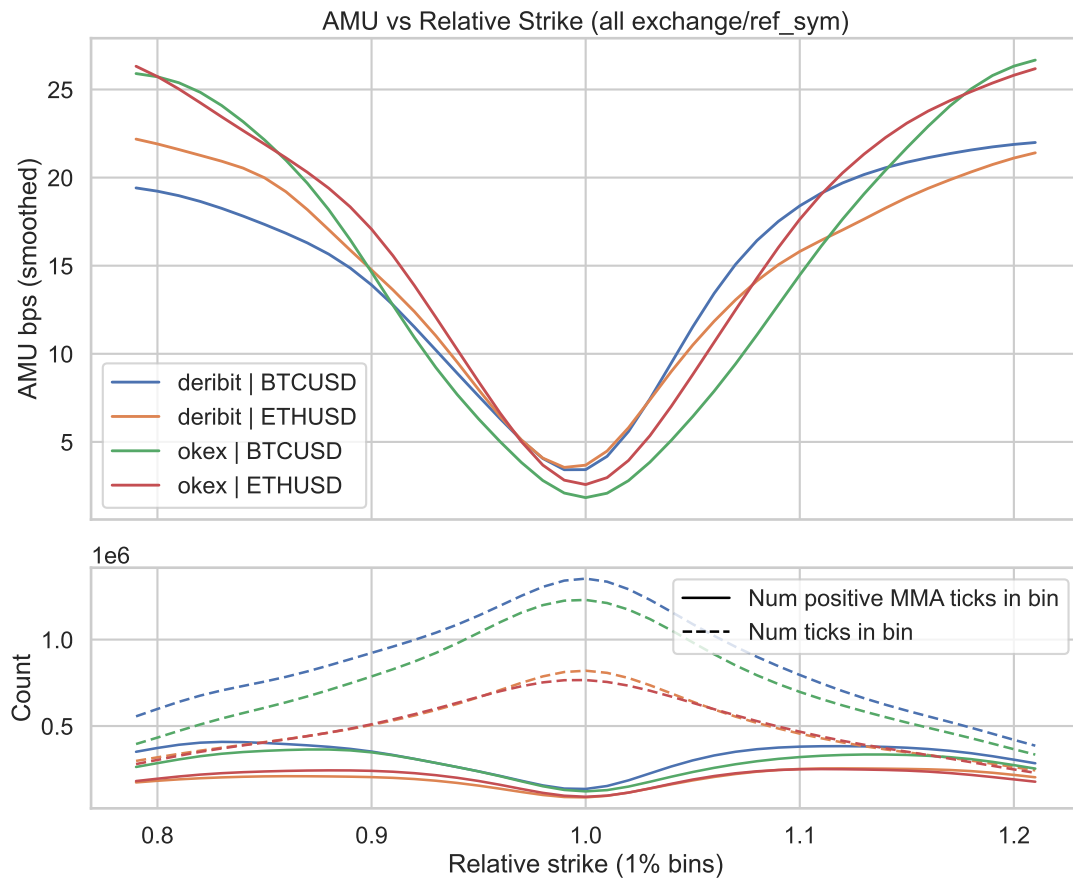


Figure 6: AMU by relative strike across exchanges and underlyings.

$$|X_{i,t}| \leq B_{i,t}, \quad (3)$$

where the band $B_{i,t}$ includes observable and latent implementation costs:

$$B_{i,t} = \tau_{i,t} + s_{i,t} + m_{i,t} + f_{i,t} + \ell_{i,t} + \kappa_{i,t}. \quad (4)$$

Here $\tau_{i,t}$ denotes explicit fees, $s_{i,t}$ bid-ask and market-impact costs, $m_{i,t}$ margin and collateral costs, $f_{i,t}$ funding and stablecoin-conversion costs, $\ell_{i,t}$ legal and regulatory uncertainty, and $\kappa_{i,t}$ counterparty, custody, and operational costs. The key empirical claim is that the post-2024 market experienced a decline in several components of $B_{i,t}$.

A reduced-form representation is:

$$B_{i,t} = \alpha_0 + \alpha_1 \text{Spread}_{i,t} + \alpha_2 \text{Depth}_{i,t}^{-1} + \alpha_3 \text{Funding}_t + \alpha_4 \text{Stress}_t + \alpha_5 \text{Regime}_t + u_{i,t}, \quad (5)$$

where Regime_t is a post-2024 regime indicator or a sequence of institutional event dummies. A negative α_5 is the empirical signature of liquidity-premium compression.

5 A threshold model of parity enforcement

The simplest dynamic model treats parity deviations as persistent inside the arbitrage band and rapidly mean-reverting outside it:

$$X_{i,t+1} = \begin{cases} \rho_0 X_{i,t} + \varepsilon_{i,t+1}, & |X_{i,t}| \leq B_{i,t}, \\ \rho_1 X_{i,t} + \varepsilon_{i,t+1}, & |X_{i,t}| > B_{i,t}, \end{cases} \quad |\rho_1| < |\rho_0|. \quad (6)$$

Inside the band, arbitrageurs do not trade aggressively because the wedge is too small after implementation costs. Outside the band, arbitrage capital enters and compresses the deviation. A market-structure transition can then be represented as a break in the threshold:

$$B_{i,t} = B_i^{\text{pre}} \mathbf{1}_{t < \tau} + B_i^{\text{post}} \mathbf{1}_{t \geq \tau}, \quad B_i^{\text{post}} < B_i^{\text{pre}}. \quad (7)$$

This model has two testable implications. First, the distribution of $|X_{i,t}|$ and AMU should shift left after the institutionalization events. Second, conditional on large deviations, mean reversion should become faster because more capital can be deployed to enforce parity.

A more structural but still tractable version lets the band depend on arbitrage capital:

$$B_{i,t} = \frac{\phi_{i,t}}{A_t}, \quad (8)$$

where $\phi_{i,t}$ is the per-unit implementation cost and A_t is effective arbitrage capacity. Regulatory clarity and ETF-linked market integration lower $\phi_{i,t}$ and increase A_t , both of which narrow the no-arbitrage band. This formulation captures the central mechanism in one sentence: the collapse of PCP violations reflects a transition from visible but non-bankable arbitrage to institutionally enforceable arbitrage.

6 Data and empirical design

6.1 Core data

The core sample uses Deribit and OKX option and futures order-book data from Tardis from 2020 through February 2026. The same AMU calculation is applied consistently across the full horizon so that pre-2024 and post-2024 comparisons are internally coherent.

For each exchange, underlying, strike, maturity, and timestamp, the dataset contains synchronized best bid and ask quotes for calls, puts, and matching expiry futures. For inverse products, all legs are converted into a common currency using the platform-specific payoff and premium conventions. The empirical sample is filtered to matched put–call pairs with the same strike and maturity, positive bid quotes, and a futures contract or forward proxy with matching expiry.

6.2 Liquidity and friction variables

The panel is grouped at the (day, exchange, ref_sym, rel_strike_bucket, tte_bucket) level and includes directly usable MMA outcomes and liquidity proxies. The primary dependent variable is the panel-cell mean MMA:

$$\overline{\text{MMA}}_{g,t} = \text{mean_mma_bp}_{g,t} \quad (9)$$

The core controls already in the panel are option spread measures (mean_call_spread_bp, mean_put_spread_bp), quote-size and depth proxies (mean_min_quote_size_dollar, bid/ask amounts), moneyness and maturity buckets (rel_strike_bucket, tte_bucket), open interest, and stale-quote fractions (frac_call_stale, frac_put_stale, frac_spot_stale).

This variable set is sufficient to test the liquidity-premium mechanism in the spirit of limits-to-arbitrage models (Amihud & Mendelson 1986, Shleifer & Vishny 1997, Gârleanu & Pedersen 2011): higher implementation frictions should map into higher average AMU. The design deliberately emphasizes variables that are observable at the quote and panel level rather than latent dealer-balance-sheet objects. The paper therefore tests whether AMU behaves as a microstructure-based proxy for the cost of parity enforcement.

In the regressions, the spread and depth controls are defined explicitly as

$$\text{Spr}_{g,t} = \frac{1}{2} \left(\text{mean_call_spread_bp}_{g,t} + \text{mean_put_spread_bp}_{g,t} \right), \quad (10)$$

$$\text{Depth}_{g,t} = \log \left(\text{mean_min_quote_size_dollar}_{g,t} \right). \quad (11)$$

Table 4 summarizes the core data-processing assumptions used to construct MMA/AMU and the regression covariates.

6.3 Institutional event dates

The institutional event dates are January 10, 2024 (U.S. spot Bitcoin ETP approval), May/July 2024 (spot Ether ETF approval/trading milestones), June 30, 2024 (MiCA stablecoin provisions), and December 30, 2024 (broader MiCA CASP framework). Consistent with the literature on institutional frictions and arbitrage capacity, these dates are treated as regime dates rather than as single-cause shocks.

In the baseline implementation, the event design is introduced through calendar dummies matched to panel day: a post-2024 regime indicator and event-window indicators

Table 4: Data-processing assumptions for MMA/AMU construction

Component	Assumption
Trading-cost parameter	<code>cost_per_notional</code> = 0.003. The cost term is <code>cost_per_notional</code> multiplied by index level.
AMU clipping rule	MMA series are clipped to the interval $[0, 100]$ basis points before positive-part averaging.
MMA clipping rule	MMA series are clipped to the interval $[-500, +500]$ basis points.
Spread filter	Option spread filters use a maximum of 1000 basis points for both call and put spreads.
Relative-strike filter	Allowed range is $[0.79, 1.21]$.
Minimum quote size	<code>min_quote_size_dollar</code> must be at least 5000.
Subsampling	Last-known observation carried forward to the configured interval (default: 5min). If the last-known bid/ask price/size observation is missing, or has not been updated in the interval the value is marked stale.

around the main institutional event dates. This keeps the identifying structure transparent. Additional controls such as ETF flows, stablecoin basis measures, and funding variables are best treated as robustness extensions rather than prerequisites for the main specification.

6.4 Empirical specifications

The empirical strategy is organized around two questions. First, does mean MMA compress after 2024 even after controlling for within-cell liquidity conditions? Second, is any post-2024 compression stronger in the segments that are historically least liquid?

The baseline panel specification is:

$$\overline{\text{MMA}}_{g,t} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{Spr}_{g,t} + \beta_3 \text{Depth}_{g,t} + \beta_4 \text{Stale}_{g,t} + \gamma_g + \varepsilon_{g,t}, \quad (12)$$

where g indexes the panel cell (exchange, ref_sym, rel_strike_bucket, tte_bucket) and γ_g are cell fixed effects. In this within-cell design, a negative β_1 after friction controls is evidence of residual post-2024 compression beyond ordinary within-cell changes in spreads, depth, and quote quality. To connect the time-series regime shift to the cross-sectional liquidity facts, an interaction specification is also estimated:

$$\overline{\text{MMA}}_{g,t} = \beta_0 + \beta_1 (\text{Post}_t \times \text{ETH}_g) + \beta_2 (\text{Post}_t \times \text{ATM}_g) + \beta_3 (\text{Post}_t \times \text{NearExp}_g) + \gamma_g + \delta_t + \varepsilon_{g,t}. \quad (13)$$

Table 5 defines the abbreviated regressor names used in Equations (10)–(13).

Here γ_g absorb the time-invariant ETH, moneyness, and maturity segment main effects, while δ_t absorb the common post-2024 level shift. The interaction coefficients β_1 – β_3 identify which segments experience stronger post-2024 compression, conditional on the fixed effects and friction controls.

Table 6 gives a compact overview of the baseline, AMU, and MMA-std specifications, and Table 7 reports the coefficient estimates with HC1 standard errors and t-statistics. Figure 7 visualizes the same estimates with 95% confidence intervals.

Table 5: Abbreviations used in empirical specifications

Abbreviation	Definition
Post_t	Post-2024 regime indicator (1 for dates on/after 2024-01-10, 0 otherwise).
$\text{Spr}_{g,t}$	Average put-call spread in basis points: $0.5 \times (\text{mean_call_spread_bp} + \text{mean_put_spread_bp})$.
$\text{Depth}_{g,t}$	Log quote-depth proxy: $\log(\text{mean_min_quote_size_dollar})$, where min quote size is taken over put and call bid/ask quote sizes.
$\text{Stale}_{g,t}$	Mean stale-quote fraction across call, put, and spot legs.
ETH_g	ETH-underlying segment indicator.
ATM_g	At-the-money segment indicator: $ \text{rel_strike_bucket} - 1 \leq 0.05$.
NearExp_g	Near-expiry segment indicator: $\text{tte_bucket} \leq 0.35$ years.
γ_g	Panel-cell fixed effect for cell g (absorbs time-invariant cell heterogeneity).
δ_t	Date fixed effect (absorbs common day-level shocks and aggregate time variation).
$\varepsilon_{g,t}$	Idiosyncratic error term for cell g at date t .

	dependent	nobs	r2
specification			
MMA	MMA	130,621	1.000
AMU	AMU	130,621	0.213
MMA Std	Std MMA	130,621	0.471

Table 6: Regression specification summary.

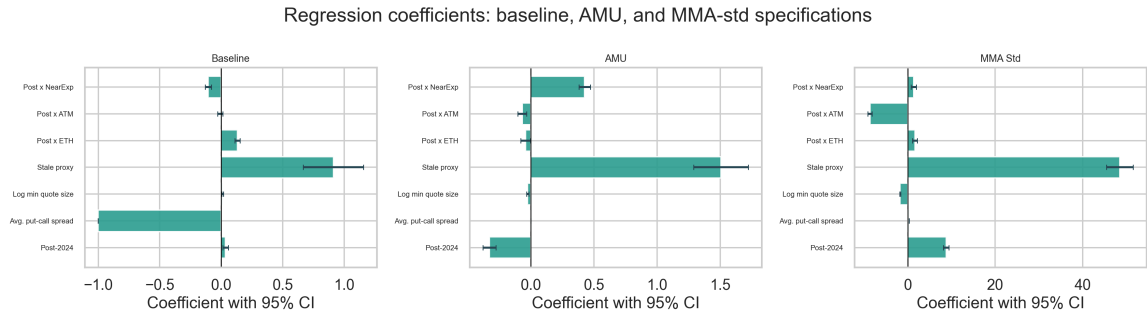


Figure 7: Coefficient plots for the panel regressions. The intervals are approximate 95% confidence intervals based on HC1 standard errors.

Term	MMA Coef	AMU Coef	MMA Std Coef	MMA $\beta \text{std}(X)/\text{std}(Y)$	AMU $\beta \text{std}(X)/\text{std}(Y)$	MMA Std $\beta \text{std}(X)/\text{std}(Y)$
Post_t	-0.027 (0.026)	-0.226 (0.142)	3.790* (2.177)		-2%	1%
$\text{Spr}_{g,t}$	-1.001*** (0.000)	-0.001*** (0.000)	0.280*** (0.004)	-100%	-4%	57%
$\text{Depth}_{g,t}$	-0.005 (0.009)	-0.025** (0.010)	-1.313*** (0.158)		-2%	-6%
$\text{Stale}_{g,t}$	0.758*** (0.200)	1.014*** (0.169)	24.751*** (2.407)		3%	4%
$\text{Post}_t \times \text{ETH}_g$	-0.169*** (0.025)	-0.588*** (0.156)	-20.842*** (2.534)		-14%	-26%
$\text{Post}_t \times \text{ATM}_g$	0.022 (0.025)	0.492*** (0.102)	9.260*** (1.895)		11%	10%
$\text{Post}_t \times \text{NearExp}_g$	0.137*** (0.029)	-0.066 (0.140)	-10.026*** (2.265)		-2%	-12%
Total R^2	1.000	0.213	0.471			
Observations	130,621	130,621	130,621			
Fixed effects	cell_id	cell_id	cell_id			

Table 7: Regression coefficient estimates with HC1 standard errors and t-statistics. Stars indicate p -value bands based on the t-statistic. In the fixed-effects rows, `cell_id` (and the subscript g) denotes the tuple (`exchange`, `ref_symbol`, `rel_strike_bucket`, `tte_bucket`).

The threshold model in Section 4 motivates the interpretation of these panel results. Once the baseline reduced-form patterns are established, threshold and local-projection designs can be used as robustness checks for whether large deviations decay faster after 2024. They are therefore complements to, not substitutes for, the fixed-effects panel evidence.

6.5 Figure roadmap

The figure set should focus on outputs generated from the 2020–2026 AMU panel:

Figure 1: Daily AMU time series by exchange and underlying (four lines: BT-CUSD/ETHUSD $\times$ Deribit/OKX), with vertical lines marking the main 2024 institutional event dates.

Figure 2: Heatmap of mean AMU over (`rel_strike_bucket`, `tte_bucket`), separately for pre-2024 and post-2024 periods.

Figure 3: Event-study plot of AMU around each institutional event date (stacked windows, e.g. $[-60, +60]$ days), shown separately by exchange.

Figure 4: Binned-scatter or partial-residual plot of MMA versus average put-call spread (bp) and log mean minimum quote size (USD), demonstrating the friction-gradient predicted by limits-to-arbitrage theory.

Figure 5: Compression decomposition plot: four separate bar charts (shared y-axis) for pre- to post-2024 AMU changes across BTC/ETH, Deribit/OKX, ATM/Non-ATM, and NearExp/FarExp.

Figure 6: Regression coefficient plot showing the baseline and interaction specifications with 95% confidence intervals.

These figures are aligned with the liquidity-premium framing while staying within the observed variables of the dataset.

For completeness, external covariates such as ETF flows, stablecoin basis, and funding

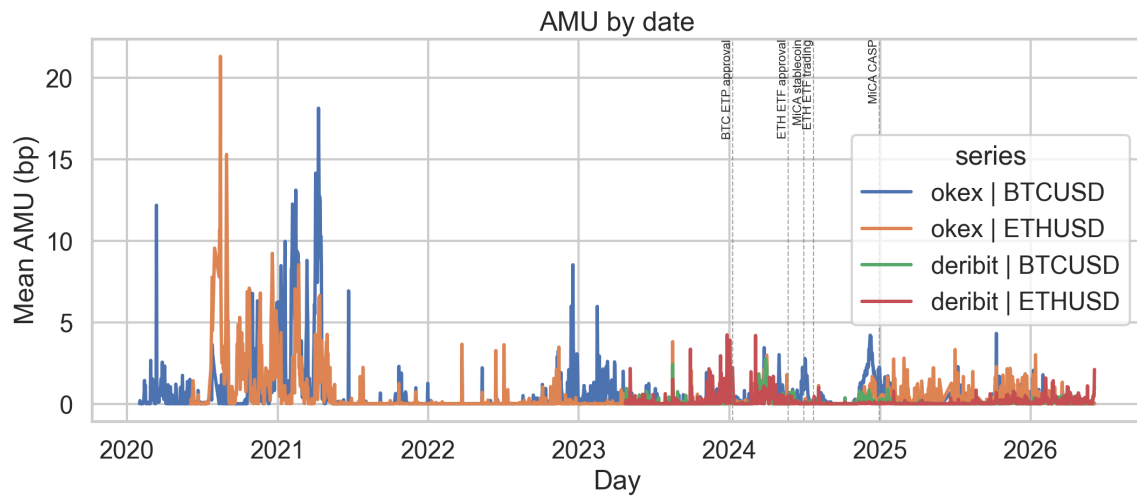


Figure 8: AMU by date, with the main 2024 institutional event dates marked.

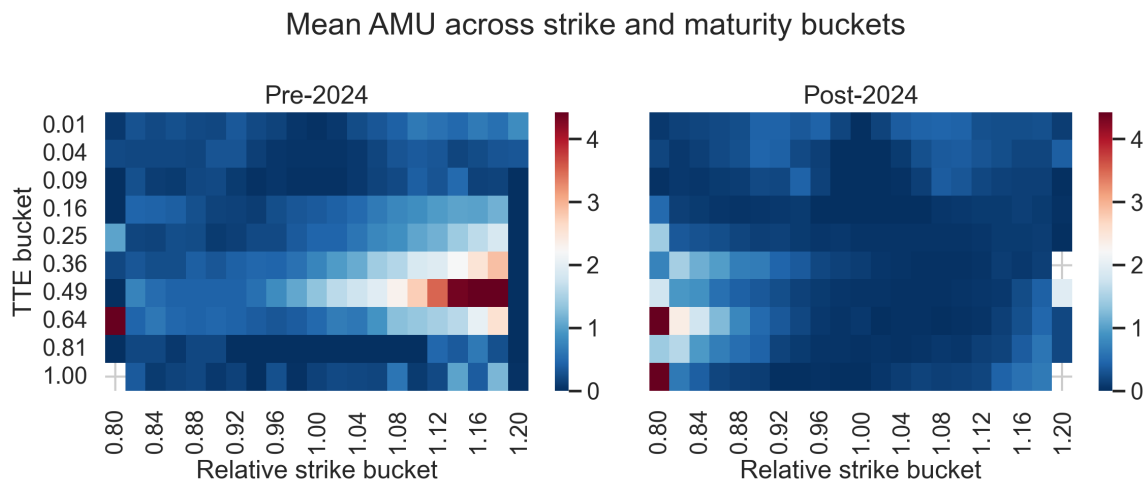


Figure 9: Mean AMU across relative-strike and maturity buckets, split into pre-2024 and post-2024 periods.

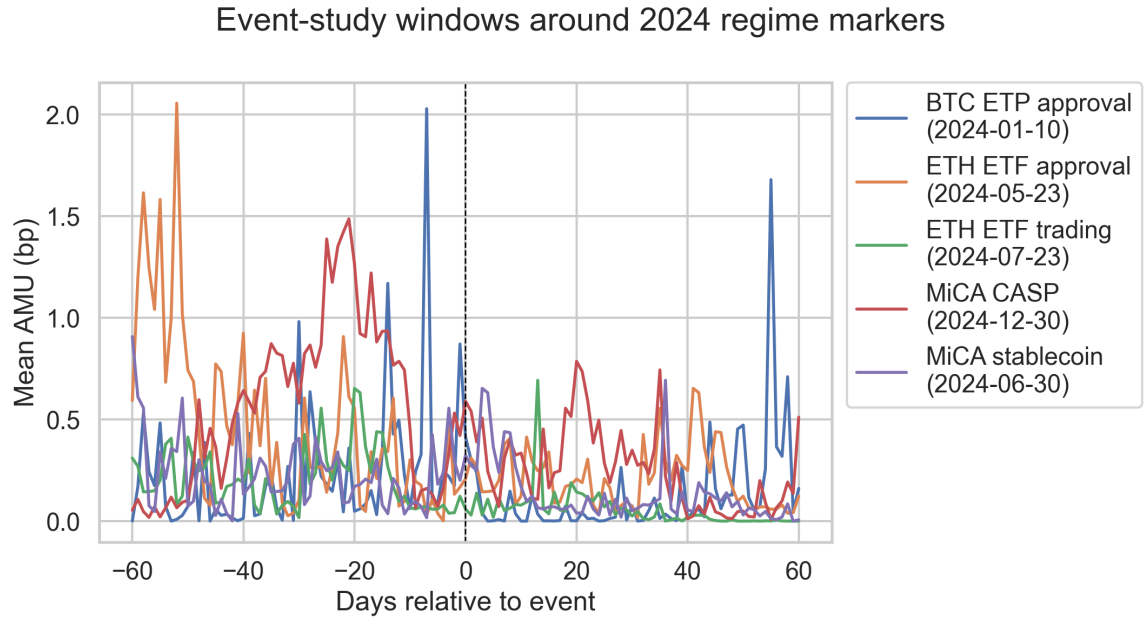


Figure 10: Event-study windows for AMU around the main 2024 institutional event dates, shown separately by exchange.

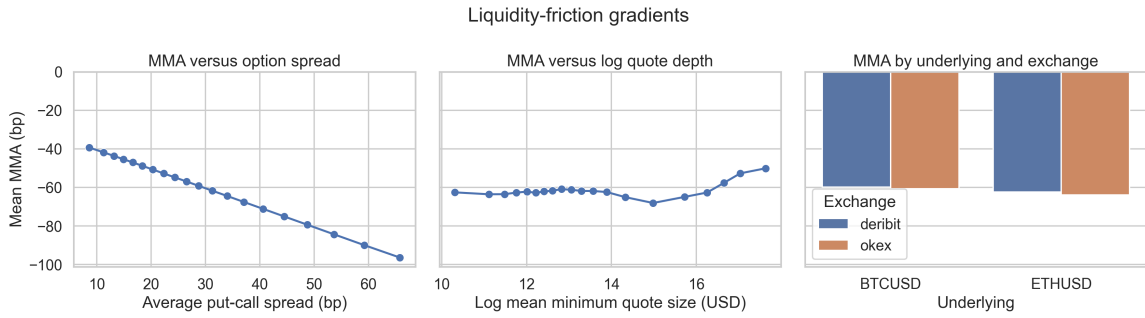


Figure 11: Global mean MMA vs put-call spread, quote size (USD) and exchange/underlying.

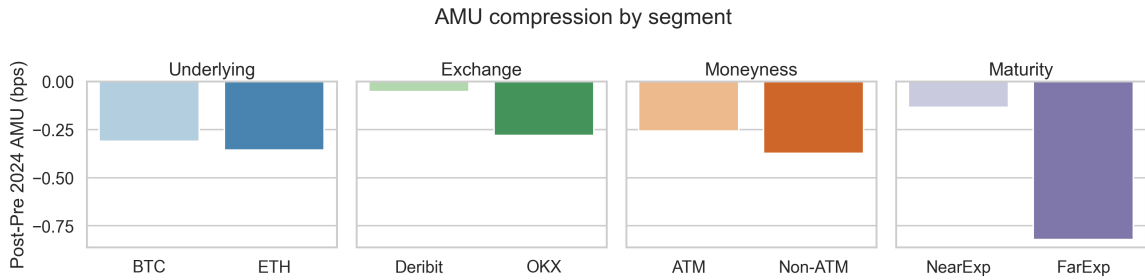


Figure 12: Pre- versus post-2024 AMU compression shown in four separate bar charts with a shared y-axis: BTC/ETH, Deribit/OKX, ATM/Non-ATM, and NearExp/FarExp.

variables can be added in a separate robustness subsection. Local projections can then test whether large deviations decay faster after 2024 in the full panel:

$$|X_{i,t+h}| - |X_{i,t}| = \theta_h |X_{i,t}| + \lambda_h (|X_{i,t}| \times \text{Post2024}_t) + \Gamma_h Z_{i,t} + \varepsilon_{i,t+h}. \quad (14)$$

A negative λ_h indicates faster post-2024 convergence.

7 Historical analogies

The crypto liquidity-premium story is best understood as part of a broader class of arbitrage relationships that are exact in theory but costly in practice. Covered interest parity is the canonical example. After the global financial crisis, CIP deviations became persistent because bank balance sheets and dollar funding constraints limited arbitrage (Du et al. 2018). In crypto options, the sign of the historical transition may be reversed: instead of new constraints widening the band, institutionalization and regulatory clarity narrowed it.

ETF arbitrage provides a second analogy. ETF premiums and discounts are controlled by authorized participants who create and redeem shares. However, when the underlying assets are illiquid or arbitrageurs face balance-sheet constraints, ETF premiums and discounts can persist (Pan & Zeng 2017, Todorov 2021). The Bitcoin ETF approvals matter in this context not only because they attracted flows, but because they inserted Bitcoin exposure into a familiar creation-redemption, custody, and brokerage architecture.

The on-the-run/off-the-run Treasury spread is a third analogy. The spread between newly issued and older Treasury securities has often been interpreted as a liquidity premium. Convergence trades can narrow the spread, but only when arbitrageurs have sufficient funding and balance sheet capacity. Similarly, the CDS–bond basis measures a deviation between synthetic and cash credit exposure and is known to vary with funding, margin, and capital costs (Gârleanu & Pedersen 2011, Duffie 2010).

These analogies suggest that the relevant object is not the existence of a no-arbitrage formula, but the institutional mechanism that enforces it. The AMU measure is useful precisely because it measures the quoted compensation left to the marginal liquidity supplier when that mechanism is incomplete.

8 Conclusion

This paper introduces Absolute Market Unfairness (AMU) as a model-free way to measure executable put–call-parity wedges in inverse cryptocurrency option markets after the relevant contract conversions and trading costs are imposed. Using Deribit and OKX BTCUSD and ETHUSD options from 2020 through early 2026, we show how AMU can be calculated consistently over time and across the option surface, and we document that the observed wedge is largest precisely where arbitrage is hardest to deploy.

The empirical value of the AMU framework is that it moves from descriptive parity violations to a market-quality measure with economic interpretation. AMU is not simply a statement that quotes differ from frictionless theory. It is a statement about how much compensation remains available to liquidity suppliers after the full arbitrage cycle is priced correctly. That makes it useful both for comparing option-market quality across venues and for studying how regulation, custody, capital access, and exchange competition affect the enforcement of the law of one price.

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